Welington Santos

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **8+** years across sports commerce, e-commerce retail, and healthcare IT, building scalable data pipelines and analytics foundations. Proficient in **dbt**, **Apache Spark**, and **Azure Fabric** with expertise in ETL orchestration, data modeling, and cloud infrastructure. Experienced designing layered warehouse architectures, implementing data quality frameworks, and operationalizing production pipelines to drive business intelligence.

SKILLS

Languages: Python, SQL, Bash, JavaScript, TypeScript, HTML, CSS

Data & ML: Apache Spark, PySpark, Pandas, NumPy, Airflow, dbt, Databricks, MLflow, TensorFlow

Database: Azure SQL, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Testing: Pytest, Great Expectations, Jest, Cypress

Cloud & DevOps: Microsoft Azure, Microsoft Fabric, OneLake, Azure Data Factory, Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, AWS, S3, CloudFront, EC2, Lambda

Others: RESTful API, GraphQL, WebSocket, Git, GitHub, Jira, Confluence, OpenAI API, Datadog, Grafana, Prometheus, Agile, Scrum

EXPERIENCE

Fanatics

Oct 2023–Present

Senior Data Engineer

- Led the design and implementation of a centralized event streaming platform using **Apache Spark** and **Kafka**, processing 500k+ daily events across sports commerce transactions and collectibles inventory, establishing the foundation for real-time analytics.
- Built end-to-end data transformation pipelines in **dbt** with comprehensive testing and documentation, normalizing transactional data from multiple payment processors and inventory systems into a curated analytics layer serving 12 analytics teams.
- Architected a multi-layer data warehouse (Bronze, Silver, Gold) using **Azure Fabric** and **OneLake**, reducing query latency from 8 minutes to under 30 seconds for ad-hoc sports fan engagement reports.
- Implemented **Great Expectations** data quality validation framework across ingestion pipelines, distinguishing warning-level anomalies from blocking failures, which reduced silent data issues reaching analytics by **87%**.

- Orchestrated complex workflow dependencies using **Azure Data Factory** pipelines with retry logic and alerting, managing incremental loads for 15+ upstream data sources and reducing failed jobs from 6 per week to 1.
- Collaborated with the analytics team to redesign the **Azure SQL** schema for fan segmentation tables—adding partitioning on event date and indexed lookups—cutting segment query times from 45 seconds to under 3 seconds.
- Established CI/CD and version control discipline for the data platform using **GitHub** and **GitHub Actions**, automating dbt model testing and deployment across development, staging, and production environments.
- Monitored and optimized pipeline performance using **Datadog** and **Azure** native cost tracking, identifying wasteful Spark shuffles and reducing monthly compute spend by **22%** while maintaining SLA compliance.
- Mentored two junior data engineers on production pipeline operations, code review best practices, and debugging distributed transformation failures in **PySpark** notebooks.
- Partnered with the data architecture team to identify and communicate early design concerns—raising questions about handling late-arriving fact data and slowly changing dimensions before implementation began.
- Supported day-to-day pipeline operations, responding to alerts and resolving blocked jobs within 1 hour, with a 99.2% on-time delivery rate for critical nightly ETL cycles.
- Implemented **Python** automation scripts alongside **Spark** to validate schema changes before promotion, reducing manual testing effort and preventing two potential production issues.

Walmart Sep 2022–Oct 2023
Senior Software Engineer

- Drove the consolidation of fragmented e-commerce transaction logs into a unified data warehouse using **dbt** and **Azure SQL**, enabling the ad-tech team to analyze 2+ billion monthly transactions for campaign optimization.
- Implemented data ingestion pipelines using **PySpark** and **Azure Data Factory** to extract semi-structured order and inventory data from legacy retail systems, processing petabytes of historical data with fault tolerance and idempotent transformations.
- Built the canonical data model for retail transactions across Walmart.com and Sam's Club using **Azure Fabric** lakehouse architecture, establishing Silver and Gold layers that improved query performance by **65%** for downstream reporting.
- Designed and deployed data quality frameworks using **Great Expectations**, creating tests for referential integrity and business rule violations that caught schema drift before it impacted dashboards.
- Optimized **Azure SQL** queries through index tuning, query plan analysis, and partitioning strategies, reducing average report execution time from 6 minutes to 1.5 minutes for the membership analytics dashboard.
- Collaborated with DataOps and FinTech teams to orchestrate interdependent ETL workflows using **Airflow**, managing SLAs for payment reconciliation pipelines that process **$500M+** in daily transactions.

- Established observability and alerting practices using **Prometheus** and custom **Python** monitoring scripts, reducing mean time to incident detection from 45 minutes to 8 minutes.
- Implemented version control and CI/CD for data transformation code using **GitHub** and **Jenkins**, enabling safe dbt model deployments across three environments with automated lineage validation.
- Mentored cross-functional teams on data engineering best practices, writing internal documentation on partition strategies and incremental load patterns that became the standard for the organization.

E-POT Jun 2018–July 2022

Full Stack Engineer

- Built real-time data pipelines ingesting healthcare system events—patient records, billing transactions, lab results—using **Apache Spark** and **Kafka**, handling 50k+ daily events with exactly-once guarantees.
- Designed and implemented a multi-tier data warehouse for healthcare analytics using **PostgreSQL** and **dbt**, creating normalized dimensional models for patient outcomes and operational dashboards serving 6 hospital networks.
- Developed **Python** and **PySpark** notebooks to extract and transform semi-structured medical data from legacy HL7 feeds and EDI files, with human review workflows for sensitive fields like diagnoses and medications.
- Managed data pipeline orchestration using **Airflow**, scheduling incremental loads with dependency chains and alerting on failures to the on-call data engineer, reducing manual intervention by **70%**.
- Implemented data quality checks using custom **Python** validation scripts and later migrated to **Great Expectations**, catching malformed records before they corrupted downstream reporting tables.
- Optimized **PostgreSQL** performance for healthcare analytics queries by adding strategic indexes on patient_id and encounter_date, reducing report generation time from 12 minutes to 2 minutes.
- Established CI/CD practices for data pipelines using **GitHub** and basic shell scripts, enabling safe promotion of Airflow DAGs and dbt models across staging and production.
- Collaborated with clinical stakeholders to document data lineage and business logic, creating data dictionaries that improved data literacy and trust in analytics among non-technical users.
- Supported on-call pipeline operations, triaging production incidents and fixing race conditions in incremental load logic that had been silently dropping records during off-peak hours.

EDUCATION

University of Tokyo Apr 2014–Mar 2018

Bachelor of Science in Computer Science